

# BrightLearn Data Analytics

## Research Assignment 1

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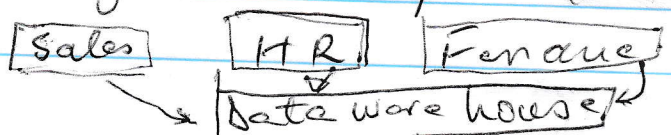
### SECTION A: DATA Fundamentals

Q1: A database is a system that is used to store information in an organised way so that people or computers can easily find, use and update that information

- Databases are important because ① they store a large amounts of information safely and ② they allow multiple users to access data at the same time
- Two real world examples are
  - ① Banking System such as where banks use databases to store eg customer details
  - ② Online shopping (eg. Amazon) where database store eg. product information

Q2 A data Warehouse - is a central system that stores large amounts of historical data from different sources, organised specifically for reporting

- While a database = daily operations, the data warehouse = analysis and decision making
- Companies use Data Warehouse to ① analyse trends over time, ② to improve decision making and to generate reports and dashboards



Reports & insights



Q3 A data lake is a place where a company stores all kinds of data, exactly as it is without organising it first.

- A data lake stores raw data, while data warehouse stores cleaned and structured data for reporting analysis
- Business prefers data lake when dealing with large volume of data, mainly coming in different formats (videos, images etc). This is important when future use of data is unknown. Examples are Social Media platforms like Facebook, where data is stored for later use.

Q4 A data mart - is a subset of a ~~data warehouse~~ data warehouse that contains data for a specific team e.g. sales, finance etc.

- Data warehouse = whole company data
- Data Mart = specific department data
- This data is used by Sales team or finance team or HR department

Q5 i) Structured data - is data that is organised in a format, usually in rows and columns making it easy to store and analyse

- Example - Tables in a database

ii) Semi-Structured Data - is a data that does not follow a strict table format but still has some structure using tags or labels - E.g. JSON

iii) Unstructured data - has no specific format and is difficult to analyse - E.g. Images or Video



Q 6: Data Modelling - is a process of designing how data is structured, stored, and related to each other in a system

- It ensures that data is organised correctly and it improves system performance and supports better decision-making
- Two types of data models are
  - ① Conceptual data Model - used for plans and understanding the systems
  - ② Logical data Model - Used for designing structure.
- Data Model = Building Plan (blue print)  
Database = Actual building

Q 7: Data Mining - is a process of analysing a large amounts of data to discover patterns, trends, and useful information

- It is used in business to understand customer behavior, to predict future trends and to personalise products and services
- Examples - ① Retail/Online shopping e.g. Amazon will use information to recommend products
- ② Banking/Fraud Detection  
Banks use data mining to detect unusual patterns of transactions, prevent fraud and to identify risky behavior

Q 8: A database Management System (DBMS) -  
- Is a system or software that allows users to create, manage, store and retrieve data in a database. - DBMS = tool that manages databases

Four Popular DBMS Platforms

① MySQL - Relational DBMS

- Stores data in tables (rows & columns)

- Eg. E-commerce websites

- Use Case - Website & applications

② PostgreSQL - Advanced Relational DBMS

- More powerful and supports complex queries

- Use Case: Data analytics and Enterprise systems

- Eg. Financial Systems

③ Oracle Database - Enterprise DBMS

- Highly secured and scalable

- Use Case - Large Organizations

- Eg. Banks and government systems

④ MongoDB - NoSQL DBMS

- Stores data in flexible formats

- Use Case: Big data and real-time applications

- Eg. Social media apps



Q9. ETL - stands for Extract, Transform, Load

- It is a process of taking raw data, prepare it for use, and storing it in a system for analysis

① Extract - Data is collected from different sources

Sources include databases, websites etc

- Eg. Sales data from different stores

② Transform - Data is cleaned and formatted

- Errors are removed

- Data is organised into a usable format

- Eg. Fixing missing values or converting currencies

③ Load - Clean data is stored in database or data warehouse

- Eg. loading data into a reporting system

Real world Example - A company like Walmart will

Extract - collect data from stores &

Transform - clean and organise the data and then

Load - store it in a data warehouse for reporting

ETL - is important - because it ensures that data is of quality and accuracy. It combines data from multiple sources and makes it ready for analysis. It helps organisations to make better decisions



Q10: Data Storage Formats - are different ways of storing and organising data so that it can be easily used, shared and analysed

① CSV (Comma-Separated Values)

- Is a format that stores data in rows and columns, separated by commas eg. Name, Age
- These are used for small datasets, Excel like data

② JSON (JavaScript Object Notation)

- It stores data in key-value pairs and is commonly used for APIs and web applications
- Eg { "Name": "John", "Age": 30 }
- This is used for APIs, Web Applications

③ Parquet - ~~Is~~ Is a Column-based storage format designed for big data processing and analytics

- Stores data by column, not rows
- Used for big data analytics, data warehouse

④ Avro - Data format used for big data that stores data together with its structure (schema)

- Used for data streaming, data exchange between systems

## SECTION B:

Q 11: AI - is when computers <sup>or machines</sup> are designed to think, learn, and make decision like humans

Examples of AI tasks include Speech recognition, Image recognition, decision-making and language translation

Two types of AI are

① Narrow AI (Weak) - Designed to perform a specific task only. It works on limited data eg. Siri and google maps navigation

② General AI (Strong AI) - A type of AI that can think, learn, and perform any intellectual task like a human. These are human level intelligence eg. future AI

Q 12: Machine Learning (ML) - is a type of Artificial Intelligence that allows computers to learn from data and improve their performance without being explicitly programmed

- Traditional programming - humans write rules and computer follows them
- Machine learning - Computers learn patterns from data and then create its own rules

Three types of ML - ① Supervised Learning  
This model learns from labelled data eg. predicting house prices or Email spam detection



- ② Unsupervised Learning - This model finds patterns in unlabelled data. Eg. Customer segmentation or grouping similar products
- ③ Reinforcement Learning - The model learns by trial and error using rewards and penalties. Eg. game playing and self driving cars

Q 13 : Deep Learning - is a type of machine learning that uses neural networks with many layers to learn from large amounts of data.

Relationship between AI, ML and Deep Learning

AI = Broad field (making machines intelligent)

ML = Subset of AI (learning from data)

Deep Learning = Subset of ML (uses neural networks)

Neural Networks - are computer systems inspired by the human brain that process data through layers of connected nodes (neurons)

Eg. Image recognition and voice assistant like Siri or Self driving cars

Q 14 : Natural Language Processing (NLP)  
- A branch of AI that enables computers to understand, interpret, and respond to human language

→ It understands text and speech. It answers questions. It translates language  
→ Eg. Chatbot (ChatGPT), Voice Assistant (Siri)  
language translation - Google Translate



Q15! A large language Model (LLM) -  
Is an advanced AI model that is trained  
on massive amounts of text data to  
understand and generate human like  
language. It can read, write and respond  
in human language

- LLM and Standard ML models differ because

① Standard ML models - Solve specific tasks  
- Use smaller datasets.  
- Limited capabilities

② LLM - Handles many language tasks  
- Trained on massive datasets and  
can generate texts, answer questions

→ Two well-known LLMs are

① ChatGPT - developed by OpenAI and used to  
answer questions, writing  
content and learning and future

② Claude - developed by Anthropic and  
used for conversations, summarizing  
documents and assisting tasks

Q16! Generative AI - is an AI that produces new  
things instead of just analysing  
data → It creates content  
- It works at high level as it is trained  
on large datasets. It learns patterns  
and relationships and it uses  
these patterns to generate new content  
- Eg. ChatGPT - Create text, Answer Q and writing  
DALL-E - Generates images from texts  
Midjourney - Creates high quality digital  
artwork



Q17 : A Training dataset - is the data that a computer learns from when building an AI model (or dataset used to teach an AI or machine learning model, how to make predictions/decisions)

- The quality of Training data is important because it determines how accurate the model is. The poor data leads to incorrect results. It helps avoid bias and errors and it improves reliability of predictions
- If the Data is Biased or Incorrect -
  - ① Biased Results - AI may favor one group over another or leads to unfair decisions. Eg. Hiring systems favouring certain gender
  - ② Incorrect Predictions - Wrong data leads to wrong outcomes Eg. Medical diagnoses errors

Q18 :- A field of AI that enables computers to see, interpret, and understand images and videos (Computer Vision = machines "seeing" and understanding images)

- How = Images are captured using cameras and then the image is converted into digital data (pixels). Then AI models analyse patterns in the image, then the system recognise objects or features
- Real-World ~~Example~~ Use Cases -
  - ① Facial Recognition - used in smartphones and security systems
  - ② Self-driving Cars - detects roads, pedestrians and traffic signs



## SECTION C

Q19 An API (Application Programming Interface) - is a set of rules that allows different software applications to communicate and share data with each other (API = bridge between Systems)

- Real World Analogy - ① Restaurant Ando
  - You (customer) → request food
  - Waiter (API) - takes request to kitchen
  - Kitchen (System) → prepares food
  - Waiter → brings food back to you

The Waiter = API

- API's - allow systems to communicate, Enable integration between applications
  - it saves time and support data sharing e.g. Payment Systems and maps and location

Q20: An API key is like a password that allows an application to use an API securely

- The purpose is to control access to an API
  - to identify who is using API
  - to track usage and limit request
  - and to improve security
- They should be stored and protected by
  - keeping it private (not publicly shared)
  - Store it securely (environment variables)
  - do not include it in public code e.g. Git Hub
  - Use encryption where possible
  - Rotate/change keys regularly



Q21 A REST API - is a type of API that uses standard web protocols (HTTP) to allow systems to communicate and exchange data

REST = Representation of State Transfer

REST API = Communication using web requests

- Main HTTP methods - ① GET - used for to retrieve data  
~~Ex~~ Eg. Get user info

② POST - Used to send/create new data. Eg. Create a new user

③ PUT - Used to update existing data  
Eg. Update user details

④ DELETE - Used to remove data  
- Eg. Delete a user

Simple Analogy - REST API = Ordering System  
→ You request (GET), add (POST), update (PUT)  
or remove (DELETE)

Q21 : JSON (JavaScript Object Notation) - is a lightweight data format used to store and exchange data between systems

JSON = data in key-value format

JSON - is commonly used because it is easy to read and write, its lightweight (small file size), it works well with API, It is supported by many programming languages and is easy for humans and machines to understand

Simple example of JSON -

```
{ "name": "John",  
  "age": 30,  
  "City": "Johannesburg" }
```

⇒ Data is stored as key: value

⇒ JSON is used in API (data exchange), Web apps, mobile apps, Configuration file

⇒ Apps like Google Maps use JSON to send location data between servers and applications

Q 23: Prompt Engineering - is how you ask questions or give instructions to AI in a clear way to get better answers

Better prompt = Better AI response

Prompt Engineering is input - to improve accuracy of AI responses. It saves time by reducing incorrect answers  
- helps AI understand the tasks clearly

Example 1 - Poor Prompt: "Tell me about business"

- Improved Prompt: "Explain the key factors for success in small retail businesses in SA in simple terms"

Example 2 - Poor Prompt: "Write something about data"

- Improved Prompt: "Write a short explanation of data analytics for beginners, including examples and a simple diagram"

A good Prompt - is clear and specific, includes context, states the desired format (eg table)  
- Avoid vague language



Q 24: A MAP function - is a programming function that applies a specific operation to each item in a list and returns a new list with the result.

MAP = apply the same operation to all items  
Simple Example = Input list:  $[1, 2, 3, 4]$

Operation: Multiply each number by 2  
Output  $[2, 4, 6, 8]$

Python Example: Numbers =  $[1, 2, 3, 4]$

~~Result = list(map(lambda x:~~

Result = list(map(lambda x:  $x*2$ , numbers))

Print(result)

Output  $[2, 4, 6, 8]$

Explanation of Code (Simple)

numbers  $\rightarrow$  input list

map()  $\rightarrow$  applies the function

lambda x:  $x*2$   $\rightarrow$  rule (multiply by 2)

result  $\rightarrow$  new list

MAP is used for  $\rightarrow$  data processing

$\rightarrow$  transforming datasets

$\rightarrow$  Cleaning and modifying data

$\rightarrow$  Data analytics tasks

$\therefore$  MAP = factory machine

each item goes through the same process

Q25: Cloud Computing - is the delivery of computing services (such as storage, servers, and software) over the internet instead of using local computers.  
Cloud = access services over the internet

Three Main Cloud Service Models

① IaaS (Infrastructure as a Service)

- Provides basic infrastructure like servers, storage and networking  
Eg Amazon Web Services (EC2)

② PaaS (Platform as a Service)

- Provides a platform for ~~development~~ <sup>and</sup> developing and running applications  
eg. Google Cloud - developers focus on building apps not infrastructure

③ SaaS (Software as a Service)

→ Provides ready to use software over the internet

- Eg Microsoft 365 → Users just use the software

Cloud Computing is important because - it reduces cost (no need for own servers)

Scalable and accessible.

Improves collaboration

Simple Analogy

IaaS = Renting land

PaaS = Renting a house

SaaS = Staying in a hotel



Q26 ! A Version Control System (VCS)

- is a tool that helps track changes made to files or code over time and allows users to manage different versions

∴ VCS = track changes + manage versions

Git - is a popular version control system used to track changes in code and collaborate with others

• It is widely used to track changes over time

- allows multiple people to work together

- Helps recover previous versions

- Supports collaboration in teams

Commit - is a saved version of your work at a specific point in time

(Snapshot of your work)

Branch - is a separate version of your ~~work~~ project where you can make changes without affecting the main version

∴ Git = track changes in a document

Commit = saving your work

Branch = working on a copy without affecting original

Question 27 - A Jupyter Notebook - is a ~~work~~ workspace where you can write code and see the results step by step along with notes and explanations.

∴ Jupyter Notebook = Code + Output + explanation in Python - is the most <sup>one place</sup> programming language commonly used

Jupyter Notebook is popular because -

- Easy to use and interactive
- Shows results immediately
- Combines Codes, text and visuals
- Great for data analysis and learning
- Supports data visualisation

In Jupyter Notebook you can

- Write and run Python code
- Analyse data
- Create charts and graphs
- Document your work
- Share results

In Real World Data Analysts use Jupyter Notebook

- Data exploration
- Machine learning experiments
- Reporting and Presentation

∴ Jupyter Notebook = digital notebook for coding and analysis



Q 28: Python - is a high-level programming language that is widely used for data analysis, AI, and software development

- ∴ Python = easy + powerful + widely used
- It is popular in Data Analytics and AI
    - = Because it is easy to learn and use
  - Large number of libraries
  - Strong support for data analysis & AI
  - Widely used in Industry and
  - Works well with big data

Four Python Libraries

- ① NumPy - Used for numerical calculation
  - Works with arrays and mathematical operations
  - eg. Calculations on large data
- ② Pandas - Used for data analysis and manipulation
  - Works with table
  - eg. Cleaning and data analysis
- ③ Matplotlib - Used for data visualization
  - Creates charts and graphs
  - eg. Bar charts, line graphs,
- ④ Scikit-learn - Used for machine learning
  - Builds predictive models
  - eg. Classification, regression

Real World Used for

- Data Analysis
- Machine learning
- Automation
- Web development

## SECTION 8

~~Q 29~~

Q 29

Data Privacy - refers to the protection of personal and sensitive information from being accessed, used or shared without permission.

∴ Data Privacy = Protection of Personal Information

→ It is important because it

- Protects information (personal)
- Prevents identity theft and fraud
- Builds trust between users and organisations
- Ensures legal compliance
- Protects sensitive business data

GDPR - The General Data Protection Regulation is a law that protects the personal data and privacy of individuals

GDPR Apply to - EU countries

- Company worldwide that protects data of EU citizen

Real World Example - Companies must protect customer data (eg. Banking details)

- If data is leaked, companies can face fines

∴ Data privacy = locking your personal information safe



Q30: Data Ethics - refers to the principles and guidelines that ensure data is collected, used, and shared in a responsible and fair way

- (= Using data responsibly and fairly)
- It is important because it -
    - protects individuals from harm
    - Prevents misuse of data
    - Ensures fairness and transparency
    - Builds trust in organisations
    - Supports legal compliance

Three Examples of Unethical Use of Data

- ① Misuse of ~~Protection~~ Personal Data
  - Eg. Using personal data without consent. Why? - it violates privacy
  - Breeds trust
  - May be illegal

→ ② Biased AI Systems

Eg. AI favouring certain groups in hiring. Why Problematic?

- Because it leads to discrimination
- Unfair outcomes
  - Ethical and legal risks

→ ③ Data Manipulations - Eg. Changing data to mislead results

Problematic because -

- it produces false insights
- leads to wrong decisions
- Damages credibility

Q31 Data governance - is the set of rules and processes used to ensure that data is accurate, secure and used properly

→ It is important because it -

→ Ensures data quality and accuracy

→ Protects data from misuse

→ Supports compliance with laws (GDPR)

→ Improves decision making

Key Components of good Data Governance Framework

① Data Quality - Ensures data is accurate and complete

② Data Security - Protects data from unauthorised access

③ Data Policies and Standards

- Rules for how data is used and managed

④ Data Ownership - Defines who is responsible for the data

⑤ Data Access Control - Determines who can view or use data

Real World Examples include A company

~~Q32~~ → Only authorised staff access sensitive data

- Data is accurate before reporting

- Policies guide how data is used



Q 32: Bias in AI refers to unfair or prejudiced outcomes produced by AI system due to biased data or flawed design (= unfair or inaccurate results)

Bias AI happens because of -

- Biased training data
- Missing or incomplete data
- Poor design of the model
- Human bias reflected in data

Real World Example → Hiring Systems  
→ A company like Amazon developed an AI hiring tool that showed bias against women - Because it was trained on past data that favored men

Potential consequences of AI Bias include

- Discrimination against certain groups
- Unfair decisions (hiring, loans)
- Loss of trust in AI systems

Q 33: Different data roles focus on different parts of working with data, from collecting and preparing data to analysing and building intelligent systems

① Data Analyst - Works with data to find insights  
- Creates reports and dashboards  
- Focus on understanding data

② Data Scientist - Builds predictive models  
- Uses machine learning  
- Focus on Predicting future outcomes

③

④

Data Engineer - Builds systems to collect and store data

- Manages data pipelines

- Focus on data infrastructure

AI Engineer - Develops AI systems

- Works with deep learning and automation

- Focus on Building intelligent systems

∴ Data Engineer - prepare data

Data Analyst - explain data

Data Scientist - predicts using data

AI Engineer - builds intelligent system

Q 34: Business Intelligence (BI) - is the process of collecting, analysing and presenting data to help organisations make better business decisions

(BI = data → insights → decisions)

Two Popular BI Tools

① Power BI - developed by Microsoft

- Used to create dashboards and reports

- Visualise data in charts and graphs

- Helps business track performance

② Tableau - Used for interactive data visualisation

- Turns data into easy to understand visuals

- Help users explore and analyse data.



BI helps business through - Supports decision making

- Identifies trend and patterns
- Improves efficiency
- Tracks performance

Real world Example include a company use BI to

- Monitor Sales performance
- Analyse customer behaviour
- Improve business strategy

## Q35 SECTION E

Q35: I found AI and Machine Learning the most interesting because it shows how computers can learn from data and make decision like humans. It is surprising to see how AI is already used in everyday life, such as recommendations on platforms like Netflix and voice assistants like Siri. This made me realise how powerful data and AI are in transforming industries and improving efficiency.

- I also found Business Intelligence (BI) the most interesting because it shows how organisations use data to make strategic decisions. It is surprising how data can be transformed into insights that improves business performance, plans and competitiveness.



### Question 361

→ I found Deep Learning and Neural Networks to be the most difficult to comprehend because the concept involves multiple layers and how machines process information similar to the human brain. At first, it was not easy to understand how the layers work together to produce output.

I still don't understand fully. I had to use online resources such as videos and the beginner platforms like Coursera and IBM and also ChatGPT.

### Question 37

→ I see data and AI playing a major role in my future career, especially in improving decision making and business performance. As someone involved in business and innovation, I believe data analytics will help me better understand customer behavior, optimize operations, and identify new opportunities. For example, in business like Agri Hub (my business), data can be used to analyse sales trends, manage inventory, and improve customer engagement. AI can also help automate processes and provide insights that support strategic planning.

In everyday life, AI is already helping through tools like Google Maps and ChatGPT, which make tasks easier and more efficient.



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